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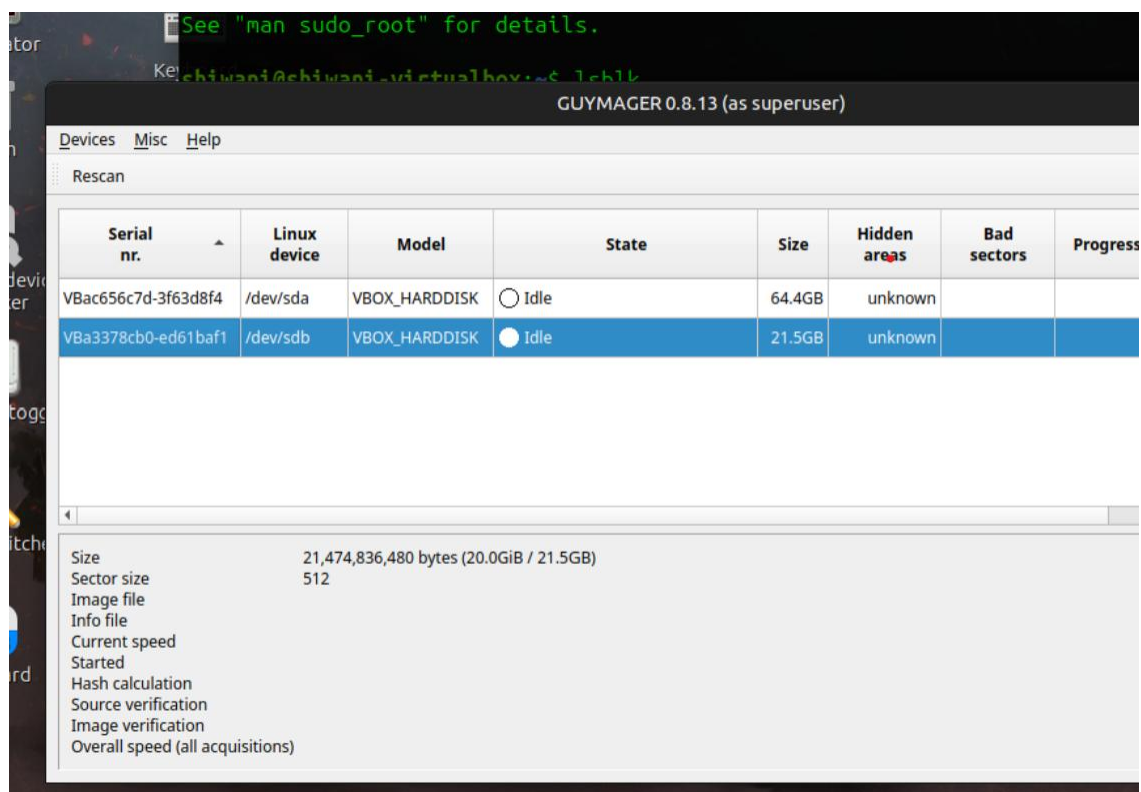
@date: May 31,2026

@imaging using guymager

- @ 19:27 : Ran lsblk in Tsurugi terminal, to confirmed the suspected disk that is sdb and it is mounted Read only.

```
shiwani@shiwani-virtualbox:~$ lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINT
sda          8:0    0   60G  0 disk 
├─sda1       8:1    0   60G  0 part /
sdb          8:16   0   20G  1 disk 
├─sdb1       8:17   0   300M  1 part 
├─sdb2       8:18   0   100M  1 part 
├─sdb3       8:19   0   128M  1 part 
├─sdb4       8:20   0   18.7G  1 part 
└─sdb5       8:21   0    780M  1 part 
sr0         11:0    1 1024M  0 rom
```

@19:28: Launched guymager and located the suspect disk that is /dev/sdb.



@ 19:29: select the suspected disk and right click and choose acquire image and make following settings.

Selected the file format as E01. Added the metadata, choose destination folder. Selected MD5 and SHA-256 hashes and enabled the verification.

Acquire image of /dev/sdb (as superuser)

File format

☐ Linux dd raw image (file extension .dd or .xxx) ☒ Split image files

☒ Expert Witness Format, sub-format Guymager (file extension .Exx) Split size 2047 MiB

Case number 101

Evidence number 1

Examiner Shiwani

Description Image Capture

Notes VBa3378cb0-ed61baf1

Destination

Image directory ... /cases/win11_guymager/

Image filename (without extension) win11_guymager

Info filename (without extension) win11_guymager_info

Hash calculation / verification

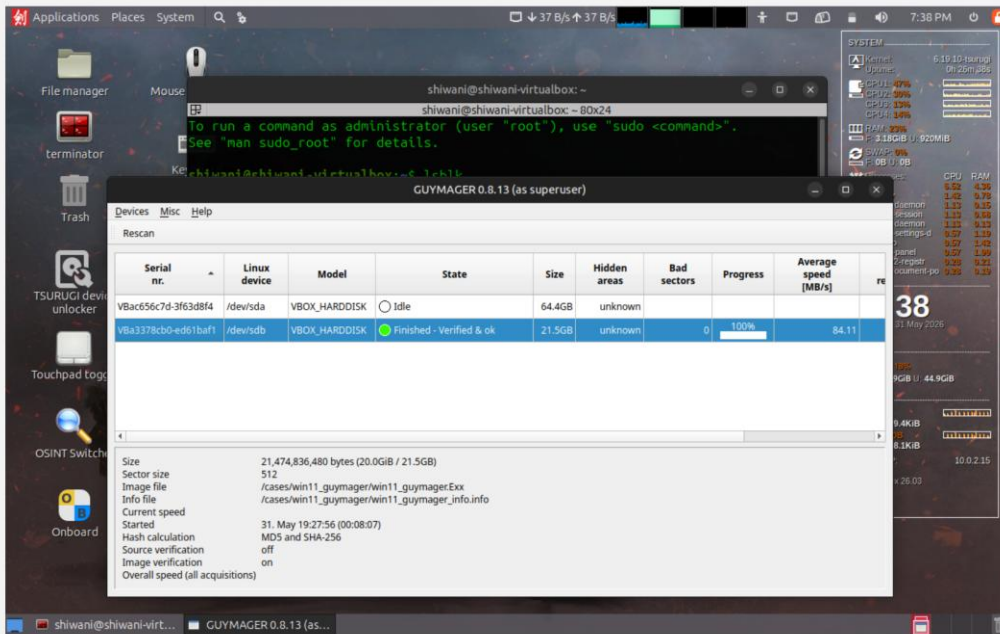
☒ Calculate MD5 ☐ Calculate SHA-1 ☒ Calculate SHA-256

☐ Re-read source after acquisition for verification (takes twice as long)

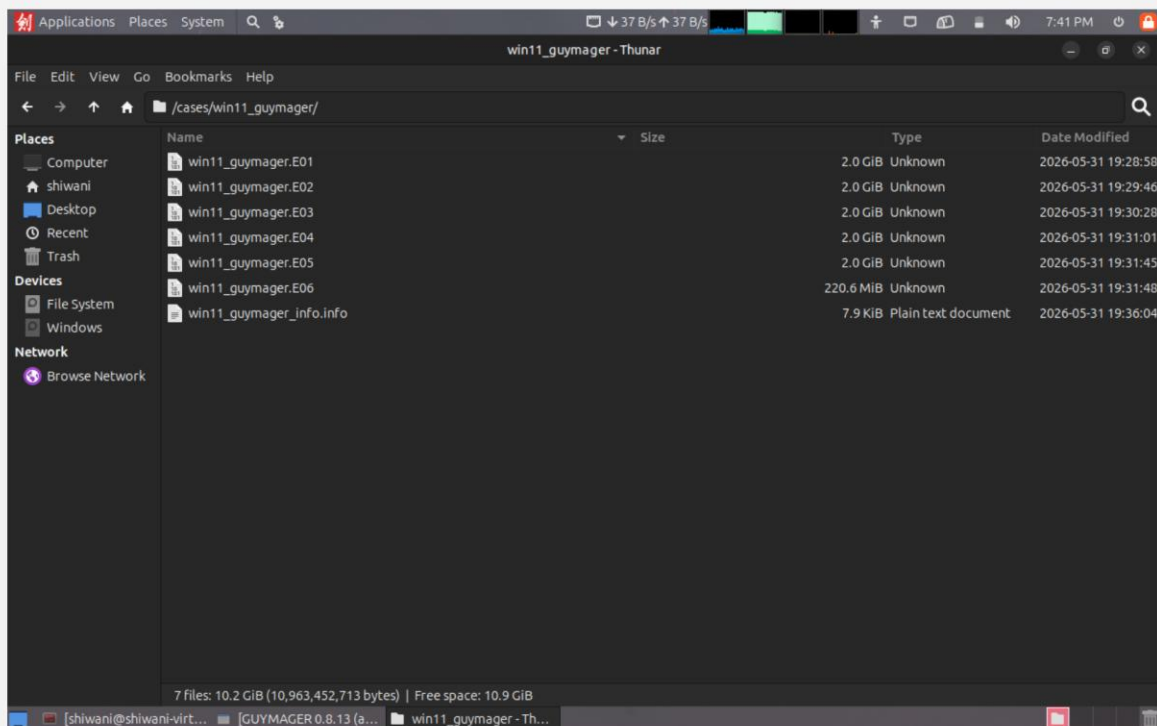
☒ Verify image after acquisition (takes twice as long)

Cancel Duplicate image... Start

@19:30: Started acquiring the image which gets finished at 19:36 with state as verified and ok.



@19:41: checked the destination folder to see all the logs related to the imaging process.



@19:42: verified the hash values from log file

MD5 hash:

Source: eba616844823e581df8413e86ede2774

Image: eba616844823e581df8413e86ede2774

- SHA-256 Hash:

Source: db6037504816d901538ea8c3bfb640485d5d1a746aaba3587714471a798acc57

Image: db6037504816d901538ea8c3bfb640485d5d1a746aaba3587714471a798acc57

Image Verification OK. The image contains exactly the data that was written proves that the integrity of data was maintained and disk was mounted as read-only.

```
win11_guymager_info.info x
120
121 No bad sectors encountered during acquisition.
122 State: Finished successfully
123
124 MD5 hash : eba616844823e581df8413e86ede2774
125 MD5 hash verified source : --
126 MD5 hash verified image : eba616844823e581df8413e86ede2774
127 SHA1 hash : --
128 SHA1 hash verified source : --
129 SHA1 hash verified image : --
130 SHA256 hash : db6037504816d901538ea8c3bfb640485d5d1a746aaba3587714471a798acc57
131 SHA256 hash verified source: --
132 SHA256 hash verified image: db6037504816d901538ea8c3bfb640485d5d1a746aaba3587714471a798acc57
133 Image verification OK. The image contains exactly the data that was written.
134
135 Acquisition started : 2026-05-31 19:27:56 (ISO format YYYY-MM-DD HH:MM:SS)
136 Verification started: 2026-05-31 19:31:47
137 Ended : 2026-05-31 19:36:04 (0 hours, 8 minutes and 7 seconds)
138 Acquisition speed : 88.66 MB/s (0 hours, 3 minutes and 51 seconds)
7: This is Geany 2.0.
```

Based on log file:

Metadata is stored properly in the .info file.

```
win11_guymager_info.info x
103 Acquisition
104 =====
105
106 Linux device : /dev/sdb
107 Device size : 21474836480 (21.5GB)
108 Format : Expert Witness Format, sub-format Guymager - file extension is .Exx
109 Image meta data
110 Case number : 101
111 Evidence number : 1
112 Examiner : Shiwani
113 Description : Image Capture
114 Notes : VBa3378cb0-ed61baf1
115 Image path and file name: /cases/win11_guymager/win11_guymager.Exx
116 Info path and file name: /cases/win11_guymager/win11_guymager_info.info
117 Hash calculation : MD5 and SHA-256
118 Source verification : off
119 Image verification : on
120
121 No bad sectors encountered during acquisition
This is Geany 2.0.
```

Hence:

Logs are properly stored in log files.

Hashes are matched and verified.

Time stamps are properly stored in the info file along with the speed.

Metadata is preserved.

```
136 Verification started: 2026-05-31 19:31:47
137 Ended : 2026-05-31 19:36:04 (0 hours, 8 minutes and 7 seconds)
138 Acquisition speed : 88.66 MByte/s (0 hours, 3 minutes and 51 seconds)
139 Verification speed : 80.00 MByte/s (0 hours, 4 minutes and 16 seconds)
140
141
142 Generated image files and their MD5 hashes
143 =====
144
145 No MD5 hashes available (configuration parameter CalcImageFileMD5 is off)
146 MD5 Image file
147 n/a win11_guymager.E01
148 n/a win11_guymager.E02
149 n/a win11_guymager.E03
150 n/a win11_guymager.E04
151 n/a win11_guymager.E05
152 n/a win11_guymager.E06
153
```

ANSWERS:

1: Guy Mager is the easiest method for imaging as it is completely GUI based and avoid the user errors as compared to the command line. Moreover, it auto calculates the hashes and auto generates the log file.

2. dc3dd uses command-line syntax which is prone to user errors, it could also lead to overwriting evidence if wrong selection was made while selecting the source or destination. In addition to this, hashing is not auto enabled in it, and if it got missed, the image will have no verification.

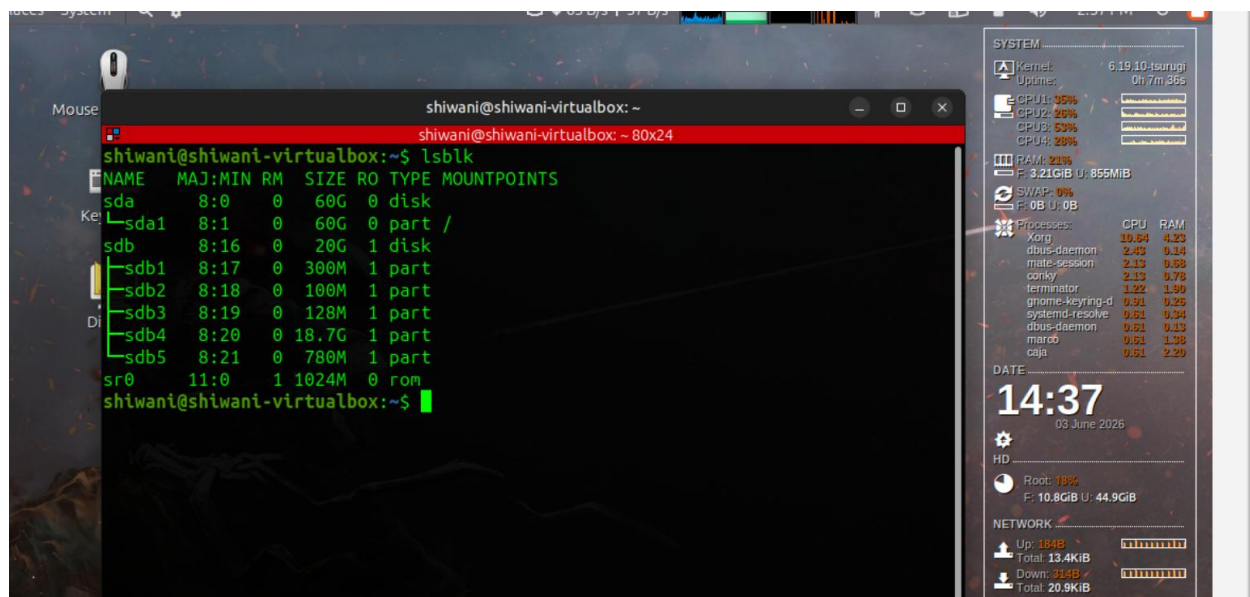
3. embedded meta data helps to explain authenticity, integrity and proper handling of evidence because it records important information such as hash values, case number, investigator name, date and time of acquisition, imaging notes. all these things help to defend the evidence. It even helps to detect the file tampering as it connects the files to specific hardware.

4 when deep-dive analysis must be done then exact sector-by-sector copy without formatting is required by investigation. Another situation is when multiple platforms have to process the data, as all forensic tools can process raw image. As it has no proprietary structure, so it is useful in long-term retention policies. When compression is not required as simplicity and transparency are important.

5 E01 imaging because it stores the case metadata which is widely accepted in courts. It supports compression to save storage space. Hash verification is built in. It is used by most forensic labs and policy agencies.

6 documentation helps to keep the integrity of evidence, as it will show when and who handled the evidence and what actions were performed. Tools help to log errors and preserve timestamps. Along with that good tools store metadata which helps to explain the authenticity.

METHOD2 imaging using dc3dd



@June 3, 2026, 15:52:50 started imaging

```
shiwani@shiwani-virtualbox: /cases/win11dd
shiwani@shiwani-virtualbox: /cases/win11dd 80x24
shiwani@shiwani-virtualbox:~$ sudo mkdir -p /cases/win11dd
shiwani@shiwani-virtualbox:~$ sudo chown shiwani /cases/win11dd
shiwani@shiwani-virtualbox:~$ cd /cases/win11dd
shiwani@shiwani-virtualbox:/cases/win11dd$ sudo dc3dd if=/dev/sdb hash=md5 hash=
sha1 hash=sha256 hash=sha512 hlog=win11hashes.txt of=win11.dd

dc3dd 7.2.646 started at 2026-06-03 15:52:50 -0400
compiled options:
command line dc3dd if=/dev/sdb hash=md5 hash=sha1 hash=sha256 hash=sha512 hlog=w
in11hashes.txt of=win11.dd
device size: 41943040 sectors (probed), 21,474,836,480 bytes
sector size: 512 bytes (probed)
21474836480 bytes ( 20 G ) copied ( 100% ), 661 s, 31 M/s

input results for device '/dev/sdb':
41943040 sectors in
0 bad sectors replaced by zeros
eba616844823e581df8413e86ede2774 (md5)
db540a8da10d2a887486eb1a8a9941eae8f9fcf8 (sha1)
db6037504816d901538ea8c3bfb640485d5d1a746aaba3587714471a798acc57 (sha256)
7706395bffb1587111897d5eb7cda56fa36cd063033e2f8971d01da44c37dc83208160b7283c
af541b0885dbd907502695ec9ed46dc0a1f8269c6d49dfa11a2 (sha512)

output results for file 'win11.dd':
```

@June 3, 2026 16:03:51 finished imaging

```
shiwani@shiwani-virtualbox: /cases/win11dd
shiwani@shiwani-virtualbox: /cases/win11dd 80x24

dc3dd 7.2.646 started at 2026-06-03 15:52:50 -0400
compiled options:
command line dc3dd if=/dev/sdb hash=md5 hash=sha1 hash=sha256 hash=sha512 hlog=w
in11hashes.txt of=win11.dd
device size: 41943040 sectors (probed), 21,474,836,480 bytes
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21474836480 bytes ( 20 G ) copied ( 100% ), 661 s, 31 M/s

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db6037504816d901538ea8c3bfb640485d5d1a746aaba3587714471a798acc57 (sha256)
7706395bffb1587111897d5eb7cda56fa36cd063033e2f8971d01da44c37dc83208160b7283c
af541b0885dbd907502695ec9ed46dc0a1f8269c6d49dfa11a2 (sha512)

output results for file 'win11.dd':
41943040 sectors out

dc3dd completed at 2026-06-03 16:03:51 -0400
shiwani@shiwani-virtualbox:/cases/win11dd$
```

Input hashes

```

shiwani@shiwani-virtualbox:/cases/win11dd$ sudo md5sum /dev/sdb
[sudo] password for shiwani:
eba616844823e581df8413e86ede2774 /dev/sdb
shiwani@shiwani-virtualbox:/cases/win11dd$ sudo shasum /dev/sdb
db540a8da10d2a887486eb1a8a9941eae8f9fcf8 /dev/sdb
shiwani@shiwani-virtualbox:/cases/win11dd$ sudo sha256sum /dev/sdb
db6037504816d901538ea8c3bfb640485d5d1a746aaba3587714471a798acc57 /dev/sdb
shiwani@shiwani-virtualbox:/cases/win11dd$ sudo sha512sum /dev/sdb
7706395bffbfe1587111897d5eb7cda56fa36cd063033e2f8971d01da44c37dc83208160b7283caf5
41b0885dbd907502695ec9ed46dc0a1f8269c6d49dfa11a2 /dev/sdb
shiwani@shiwani-virtualbox:/cases/win11dd$ █

```

Output file for hashes stored in win11hashes.txt

```

shiwani@shiwani-virtualbox:/cases/win11dd$ cat win11hashes.txt
dc3dd 7.2.646 started at 2026-06-03 15:52:50 -0400
compiled options:
command line dc3dd if=/dev/sdb hash=md5 hash=sha1 hash=sha256 hash=sha512 hlog=w
in11hashes.txt of=win11.dd er has only 1.1 GB disk space remaining.

input results for device `/dev/sdb':
  eba616844823e581df8413e86ede2774 (md5)
  db540a8da10d2a887486eb1a8a9941eae8f9fcf8 (sha1)
  db6037504816d901538ea8c3bfb640485d5d1a746aaba3587714471a798acc57 (sha256)
  7706395bffbfe1587111897d5eb7cda56fa36cd063033e2f8971d01da44c37dc83208160b7283c
af541b0885dbd907502695ec9ed46dc0a1f8269c6d49dfa11a2 (sha512)

output results for file `win11.dd':

dc3dd completed at 2026-06-03 16:03:51 -0400

shiwani@shiwani-virtualbox:/cases/win11dd$ █

```

Hash values:

Source values:

md5

eba616844823e581df8413e86ede2774

sha1

db540a8da10d2a887486eb1a8a9941eae8f9fcf8

sha256:

db6037504816d901538ea8c3bfb640485d5d1a746aaba3587714471a798acc57

sha512:

7706395bffbfe1587111897d5eb7cda56fa36cd063033e2f8971d01da44c37dc83208160b7283caf54
1b0885dbd907502695ec9ed46dc0a1f8269c6d49dfa11a2

Image values:

md5

eba616844823e581df8413e86ede2774

sha1

db540a8da10d2a887486eb1a8a9941eae8f9fcf8

sha256:

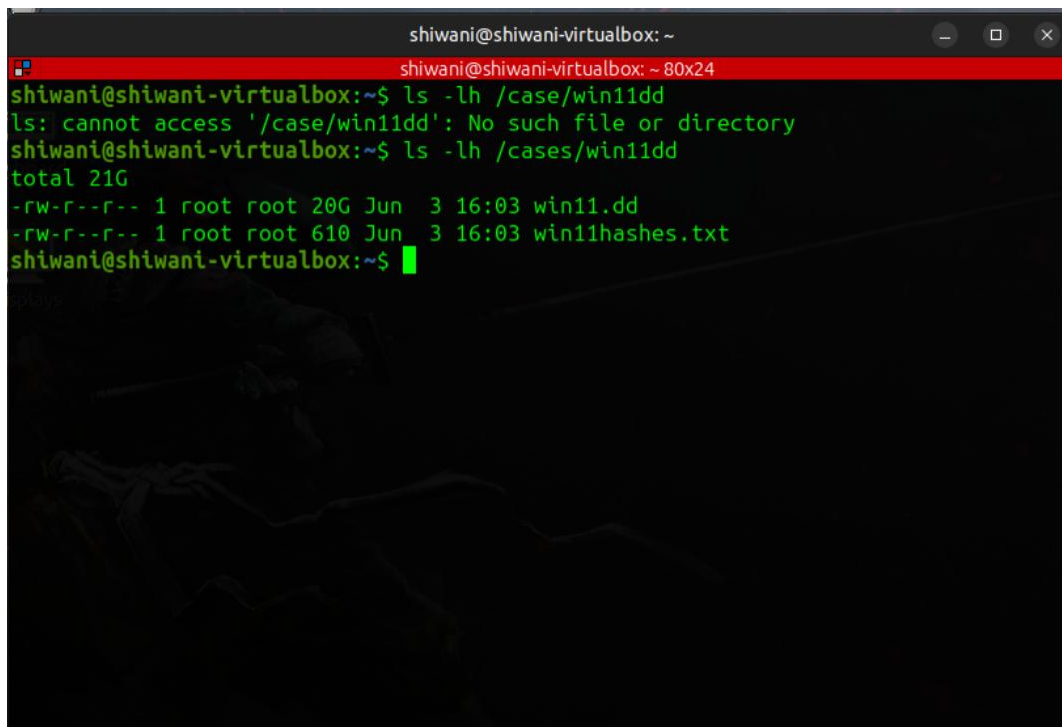
db6037504816d901538ea8c3bfb640485d5d1a746aaba3587714471a798acc57

sha512:

7706395bffb1587111897dSeb7cda56fa36cd063033e2f8971d01da44c37dc83208160b7283caf54
1b0885dbd907502695ec9ed46dc0a1f8269c6d49dfa11a2

Both are same.

Successful image

A screenshot of a terminal window titled 'shiwani@shiwani-virtualbox: ~'. The terminal shows the following commands and output:

```
shiwani@shiwani-virtualbox:~$ ls -lh /case/win11dd
ls: cannot access '/case/win11dd': No such file or directory
shiwani@shiwani-virtualbox:~$ ls -lh /cases/win11dd
total 21G
-rw-r--r-- 1 root root 20G Jun  3 16:03 win11.dd
-rw-r--r-- 1 root root 610 Jun  3 16:03 win11hashes.txt
shiwani@shiwani-virtualbox:~$
```

ewfAcquire:

@Jun3,2026 15:05 started imaging using ewfAcquire

```
shiwani@shiwani-virtualbox: /cases/win11_e01
shiwani@shiwani-virtualbox: /cases/win11_e01 80x24
shiwani@shiwani-virtualbox:~$ sudo mkdir -p /cases/win11_e01
shiwani@shiwani-virtualbox:~$ cd /cases/win11_e01
shiwani@shiwani-virtualbox:/cases/win11_e01$ sudo ewfacquire /dev/sdb
ewfacquire 20240308

Device information:
Bus type:          ATA/ATAPI
Vendor:            ATA
Model:             VBOX HARDDISK
Serial:            VBa3378cb0-ed61baf1

Storage media information:
Type:              Device
Media type:        Fixed
Media size:        21 GB (21474836480 bytes)
Bytes per sector:  512

Acquiry parameters required, please provide the necessary input
Image path and filename without extension: no extension
Case number: 1
Description: imaging win11
Evidence number: 101
Examiner name: ShiwaniJ
Notes: practicing
```

lsblk

```
shiwani@shiwani-virtualbox:/cases/win11_e01$ lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda         8:0    0  120G  0 disk
└─sda1      8:1    0   60G  0 part /
sdb         8:16   0   20G  1 disk
├─sdb1      8:17   0  300M  1 part
├─sdb2      8:18   0  100M  1 part
├─sdb3      8:19   0  128M  1 part
├─sdb4      8:20   0  18.7G  1 part
└─sdb5      8:21   0   780M  1 part
sr0         11:0    1 1024M  0 rom
shiwani@shiwani-virtualbox:/cases/win11_e01$
```

@jun 3,2026 entering the metadata.

```
shiwani@shiwani-virtualbox: /cases/win11_e01
shiwani@shiwani-virtualbox: /cases/win11_e01 80x24
Wipe sectors on read error (mimic EnCase like behavior) (yes, no) [no]: yes
The following acqiry parameters were provided:
Image path and filename:      no extension.e01
Case number:                  1
Description:                  imaging win11
Evidence number:              101
Examiner name:                ShiwaniJ
Notes:                       practicing
Media type:                   fixed disk
Is physical:                  yes
EWF file format:              original EWF (.e01)
Compression method:           deflate
Compression level:            best
Acquiry start offset:         0
Number of bytes to acquire:    0 (until end of input)
Evidence segment file size:    1.4 GiB (1493172224 bytes)
Bytes per sector:             512
Block size:                   64 sectors
Error granularity:            64 sectors
Retries on read error:        2
Zero sectors on read error:    yes
Continue acquiry with these values (yes, no) [yes]: yes
```

@jun03,2026 21:53 successfully acquired the image.

```
Status: at 99.7%
        acquired 19 GiB (21416411136 bytes) of total 20 GiB (21474836480 bytes)
        completion in 8 second(s) with 6.2 MiB/s (6581316 bytes/second)

Acquiry completed at: Jun 03, 2026 21:53:40

Written: 20 GiB (21474836668 bytes) in 54 minute(s) and 16 second(s) with 6.2 MiB/s (6595465 bytes/second)
MD5 hash calculated over data:      eba616844823e581df8413e86ede2774
ewfquire: SUCCESS
shiwani@shiwani-virtualbox: /cases/win11_e01$
```

@ Jun 03,2026 22:02:22 Verification started

```
shiwani@shiwani-virtualbox:~/cases$ ewfverify /cases/win11_e01/sha.E01
ewfverify 20240308

Verify started at: Jun 03, 2026 22:02:22
This could take a while.

Status: at 3.7%
        verified 766 MiB (804192256 bytes) of total 20 GiB (21474836480 bytes)
        completion in 1 minute(s) and 42 second(s) with 193 MiB/s (202592796 bytes/second)

Status: at 5.7%
        verified 1.1 GiB (1243152384 bytes) of total 20 GiB (21474836480 bytes)
        completion in 2 minute(s) and 10 second(s) with 148 MiB/s (155614757 bytes/second)

Status: at 7.5%
        verified 1.5 GiB (1620541440 bytes) of total 20 GiB (21474836480 bytes)
        completion in 2 minute(s) and 27 second(s) with 128 MiB/s (135061864 bytes/second)

Status: at 9.1%
        verified 1.8 GiB (1971879936 bytes) of total 20 GiB (21474836480 bytes)
        completion in 2 minute(s) and 38 second(s) with 117 MiB/s (123418600 bytes/second)
```

@ jun 3,2026 22:08:11 hashes are verified.

```
Status: at 92.8%
verified 18 GiB (19929038848 bytes) of total 20 GiB (21474836480 bytes)
completion in 25 second(s) with 57 MiB/s (60153603 bytes/second)

Status: at 93.9%
verified 18 GiB (20174307328 bytes) of total 20 GiB (21474836480 bytes)
completion in 21 second(s) with 57 MiB/s (60153603 bytes/second)

Status: at 95.0%
verified 19 GiB (20408434688 bytes) of total 20 GiB (21474836480 bytes)
completion in 17 second(s) with 57 MiB/s (60153603 bytes/second)

Status: at 96.6%
verified 19 GiB (20745093120 bytes) of total 20 GiB (21474836480 bytes)
completion in 12 second(s) with 57 MiB/s (60322574 bytes/second)

Status: at 99.1%
verified 19 GiB (21300019200 bytes) of total 20 GiB (21474836480 bytes)
completion in 2 second(s) with 58 MiB/s (61356675 bytes/second)

Verify completed at: Jun 03, 2026 22:08:11

Read: 20 GiB (21474836480 bytes) in 5 minute(s) and 49 second(s) with 58 MiB/s (61532482 bytes/second)

MD5 hash stored in file: eba616844823e581df8413e86ede2774
MD5 hash calculated over data: eba616844823e581df8413e86ede2774

ewfverify: SUCCESS
shiwani@shiwani-virtualbox:~/cases$
```